

Practical Exercise — Sorted Singly Linked List

Abstract Data Types • Java

Objective

Implement a singly linked list ADT that stores integers. Your program will generate 10 random integers, insert them into the linked list one at a time, and maintain the list in sorted order.

Requirements

1. Define a `ListNode` class containing an integer element and a reference to the next node.
2. Define a singly linked list ADT containing a reference to the first node (head).
3. Provide an operation to insert an integer into the list in its correct sorted position.
4. Generate 10 random integers.
5. Print the 10 integers before they are inserted into the list.
6. Insert the integers into the linked list.
7. Traverse the linked list and print the 10 integers in sorted order.

Example output

Numbers generated:

47 12 83 25 6 51 19 72 34 8

Numbers in sorted order:

6 8 12 19 25 34 47 51 72 83

Important

The sorted order must be produced by traversing your linked list. Do not sort the original array, vector, or other collection and print that instead.

The purpose of the exercise is to demonstrate that you can:

- create linked-list nodes;
- link nodes together;
- find the correct insertion position;
- change the next references correctly;
- traverse a linked list.

Submission

8. Your Java source code.
9. A screenshot of your program output.

The output screenshot must clearly show both the original 10 randomly generated numbers and the same numbers in sorted order after traversing the linked list.

Pair work

You may work in pairs. Both students should understand the program and be able to explain how the insertion operation works.

Before you start

Work out the insertion algorithm on paper before writing the Java code. Think carefully about the special case where the new node becomes the first node.